

Sealing Upgrade for the Enclosure: Design Summary

What this is

The original enclosure closed with screws only, with no real seal. I added a gasket seal so it can resist rain, dust and splashes. The original design wasn't touched: all changes were made on a separate copy ("v2 seal") as new steps at the end of the timeline.

The idea in one line

A soft rubber cord sits in a groove on a widened rim around the box. When the lid is screwed down flat against it, the cord is gently squeezed and blocks water and dust.

Why a wider rim

The walls are only about 2.2 mm thick, too thin to hold a groove. So I added a rim (flange) that sticks outward around the top edge of the box, and the groove is cut into that. The groove sits outside the screws, so the screw posts stay inside the sealed area and nothing had to be moved.

Rim and groove dimensions

- Rim width: 5.8 mm outward from the outer wall (1.5 mm material + 2.8 mm groove + 1.5 mm material)
- Rim thickness: 3 to 4 mm
- Groove: straight rectangular cut, 1.5 mm deep, 2.8 mm wide, no angle or round profile (those are harder to print and give no benefit)
- The groove follows the box outline. With the box's 8 mm corner rounding, the groove corners come out around 11 mm radius, which is gentle enough that the cord won't kink
- The lid has a matching flat rim, so no tongue is needed

Ribs under the rim

The rim sticks out, so on a 3D printer it needs support underneath. Small triangular ribs at 45° under the rim do this, so it prints without added supports.

- 5 ribs along each long side, spaced about 30 to 35 mm
- 3 ribs along each short side, spaced about 20 mm
- Thickness: 1.2 mm
- No fillets on the ribs. They are only print supports and don't need them

Box depth

I increased the enclosure depth from 43 mm to about 47 mm using change parameters. Otherwise the button would have been too close to the rim and its ribs.

Screws and washers

- The screws stay where they were. The base has 4.3 mm holes for heat inserts and the lid has 3.3 mm clearance holes, both unchanged
- Water could still run down a screw hole from the top of the lid, so each lid screw gets a rubber sealing washer (M3 bonded seal washer)

- The lid counterbore was 6 mm wide, which is too small for a washer. It is enlarged to about 7.5 mm (washer outer diameter + 0.3 mm) and stays 1.85 mm deep. About 1.95 mm of lid remains under it, which is plenty
- Measure the real washer before finalizing the 7.5 mm
- Screw length must cover: 1.95 mm of lid + washer thickness (0.5 to 0.8 mm) + the gap to the insert + at least 4 to 5 mm of thread in the insert. Check the insert depth before ordering screws

The seal cord

- Suggested: 2 mm round rubber cord, NBR 70 shore hardness (silicone if the device gets hot or sits in sun). This size is a sensible choice for this groove, not a fixed standard, so the client can adjust it as long as the groove is resized to match
- Cut length is the groove's centerline length minus about 2%, so it sits snug without bunching in the corners. For a 180 x 85 box this comes out around 535 mm, cut to about 524 mm. Confirm the real number with Fusion's Measure tool on the groove's centerline sketch
- Joining the ends with cyanoacrylate glue is for the prototype only. A production version should use a moulded or professionally joined ring

Assembly

1. Clean the groove and remove any printing strings
2. Cut the cord square, glue the ends together and hold a few seconds
3. Press the ring into the groove
4. Place the lid, put a washer under each screw head, and tighten gradually in a criss-cross order until the lid and base meet flat
5. Test before delivery

What it handles and what it doesn't

- Rain, splashes and dust: this design works well for these
- Full immersion: harder. 3D-printed layers are slightly porous, and the glued cord joint is a weak point. For immersion, print the walls and rim solid (4+ walls, high infill), coat the surface with epoxy or similar, and pressure test the box. Any cable entry needs its own sealed gland. How deep and how long it stays underwater decides if this is enough